IEEE Robotics & Automation Society · VIT Chennai Student Chapter

HACKVERSE: INTO THE WEB

Sprint 1 – Rapid Vibe Coding

24-Hour Hackathon · VIT Chennai · 2026

PS-01 — Multi-Agent Autonomous Financial Intelligence System for Retail Investors

Problem Statement Design a multi-agent AI system that converts real-time market data, regulatory filings, and behavioral signals into explainable, personalized investment intelligence for retail investors — bridging the gap between raw financial data and actionable decision-making. The system should not just surface what is happening in the market, but reason about what it means for a specific user's financial position, and be able to justify that reasoning transparently at every step.

Context India's retail investment ecosystem does not fail because data is unavailable — NSE price feeds, SEBI corporate filings, FII flow disclosures, earnings transcripts, and options chain data are all publicly accessible. It fails in the gap between data availability and decision intelligence: converting a continuous stream of market events into guidance that is timely, personalized, explainable, and safe to act on for an investor who has no professional research infrastructure behind them.

SEBI's own 2024 data confirms that 89% of retail F&O participants in India lose money. India added 130 million new retail investors in four years, 80% of them under 30. A hedge fund deploys parallel analyst teams running simultaneous research across fundamentals, technicals, sentiment, and macro before committing to a position. A retail investor gets a price chart and a Telegram tip. This information asymmetry is not a knowledge gap — it is an infrastructure gap. No single existing tool assembles public financial data into a coordinated, reasoned, multi-perspective research output accessible to a first-time investor in under 60 seconds.

Most systems built in this space solve one slice of the problem well. Signal screeners detect price anomalies. News aggregators surface headlines. Portfolio trackers record holdings. What is rarer is the connective tissue between them — an orchestrated reasoning layer that pulls from multiple data sources simultaneously, weights outputs against a specific user's risk profile and behavioral history, produces a synthesized and cited recommendation, and can justify its logic to a human observer at any point in the session. This gap matters more as retail participation deepens into derivative and mid-cap markets, where the cost of an unexplained wrong decision is measured in real capital loss rather than a failed simulation run.

Dependencies

- A live or simulated market data source providing real-time or near-real-time price, volume, and technical indicator feeds for listed equities.
- A document corpus of regulatory and financial disclosures (e.g. SEBI filings, earnings transcripts, or equivalent synthetic documents) suitable for semantic retrieval
- A vector database or semantic search layer capable of retrieving contextually relevant document chunks given a natural language query
- A multi-agent orchestration framework capable of dispatching parallel reasoning tasks and synthesizing outputs from multiple independent agents
- A user behavioral profiling mechanism to capture risk preference, portfolio composition, and historical interaction patterns
- A visualization or interface layer capable of rendering live signal outputs, agent reasoning traces, and portfolio state in real or near-real time
- A logging and persistence mechanism to store agent outputs, user decisions, and performance metrics across sessions

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Minimum Requirements

- A signal classification module that evaluates market data across at least three independent dimensions (e.g. price momentum, volume anomaly, sentiment) and produces a classified output with a stated confidence level and cited reasoning
- A retrieval-augmented generation component that queries a document corpus and grounds at least one agent output in retrieved source material, with attribution visible to the user
- A multi-agent architecture in which at least three specialized agents execute in parallel, each with a defined role and structured output contract consumed by a synthesis layer
- A user profiling component that modifies agent outputs based on individually stored risk parameters or behavioral history, demonstrably producing different outputs for different user profiles on identical market inputs
- A live interface rendering at minimum: current market signals with classification labels, synthesized agent output with source attribution, and current user portfolio or watchlist state

- A performance log capturing at least three measurable metrics per session (e.g. signal accuracy against 30-day forward return, agent response latency, portfolio risk concentration score)
- At least one working end-to-end demo scenario from raw data ingestion through multi-agent reasoning to a user-facing recommendation, with the full reasoning chain visible
- Graceful handling of at least one degraded-data scenario (e.g. an unavailable data feed, a missing filing, or a conflicting signal from two agents) without the pipeline failing or producing an uncited output
- A brief written summary of the agent architecture and decision logic used, for judges to review alongside the demo